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Optimal ground coverage ratios for tracked, fixed-tilt, and vertical photovoltaic systems for latitudes up to 75°N

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ABSTRACT

General guidelines for determining the layout of photovoltaic (PV) arrays were historically developed for monofacial fixed-tilt systems at low-to-moderate latitudes. As the PV market progresses toward bifacial technologies, tracked systems, higher latitudes, and land-constrained areas, updated flexible and representational guidelines are required. Using our 3D view-factor PV system model, DUET, we provide formulae for ground coverage ratios (GCRs –i.e., the ratio between PV collector length and row pitch) providing 5%, 10%, and 15% shading loss as a function of mounting type and module type (bifacial vs monofacial) between 17-75°N. Fixed-tilt arrays span a wide range of GCR (0.15–0.68, 5% loss) compared to single-axis tracked arrays (0.17–0.32) and vertical east–west arrays (0.11–0.16). We additionally optimize fixed-tilt module tilt, finding that the optimum tilt can vary from 7° above latitude-tilt to 60° below latitude-tilt in certain cases. We demonstrate that tracked and fixed-tilt PV arrays should have similar GCRs >55°N, but tracked systems are more sensitive to row-to-row shading losses <55°N. The GCR of fixed-tilt arrays at lower latitudes can reach 0.55 without introducing >2.5% shading loss, whereas tracked and vertical arrays reach 2.5% shading loss by GCRs <0.22 and <0.10, respectively. We additionally find that bifacial PV arrays require GCRs up to 0.03 lower than monofacial GCRs. These results can inform future deployment designs for latitudes >15°N.

1. Introduction

The inter-row spacing of photovoltaic (PV) arrays is a major design parameter that impacts both a system's energy yield and land-use, thus affecting the economics of solar deployment. Adjacent rows in a PV array introduce energy yield loss via direct beam shading and diffuse-sky masking (Appelbaum and Aronescu, 2022; Van Schalkwijk et al., 1997) and contribute to greater irradiance inhomogeneity and current mismatch losses. These negative effects of neighbouring PV rows can be reduced by increasing the spacing between rows and by moving tracked PV off-sun during morning and afternoon hours in a loss-minimizing practice known as *backtracking*. However, as PV deployment continues to accelerate with decarbonization plans predicting unprecedented renewable sector expansion (International Technology Roadmap for Photovoltaic (ITRPV), 2021; The International Energy Agency, 2021), PV land-use may become more limited and costly (Bolinger and Bolinger, 2022; Kafka and Miller, 2020; Tawalbeh et al., 2021). Optimization of PV array configuration within a constrained field is required, and previous guidelines for PV row spacing which focus on eliminating shading may not be adequate.

A commonly cited approach for determining inter-row spacing in fixed-tilt systems is the "winter solstice rule", where spacing is determined by the shadow length cast on the winter solstice at solar noon. Some versions of this rule call for even wider row spacing by instead constraining the time with no direct shading to between the hours of 9:00 AM and 3:00 PM on the winter solstice. The winter solstice rule is not a practical method for determining row spacing when the economics of land-use, cabling cost, and associated voltage power losses must be considered (Appelbaum and Aronescu, 2022; Sanchez-Carbajal and Rodrigo, 2019).

Similarly, the general guideline for traditionally deployed equator-facing fixed-tilt system module inclination is to set the module tilt to a value between 5-15° less than the latitude (Calabro, 2013; Lewis, 1987; Qiu and Ruiffat, 2003), which is particularly important for higher latitudes (Armstrong and Hurley, 2010; Schroder, 2011; International Renewable Energy Agency, 2014). The interplay of array tilt and row spacing must also be considered (Rehman et al., 2020), as the incorrect coupling of tilt and row spacing can lead to unnecessary production loss.

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For example, a PV plant located in Qatar has reported a 4% annual loss in energy yield due to a non-ideal coupling of tilt and row spacing, described by Shah et al. (2019).

While these general guidelines provide a simple methodology for preliminary system design, greater flexibility on row spacing requirements and characterization of associated shading loss is needed to minimize the levelized cost of electricity as available land becomes more scarce and costly. Moreover, these approaches were initially developed for monofacial equator-facing fixed-tilt systems at low-to-moderate latitudes. Now that the PV market is globally shifting towards single-axis tracking and bifacial technologies (The International Energy Agency, 2021), while also expanding to higher latitudes (Frimannslund et al., 2021; Pike et al., 2021), an investigation into optimal row spacing – or *ground coverage ratio* (GCR) (i.e., the ratio between PV collector length and row pitch) – for a wider range of technologies and locations is warranted.

Among a growing configuration of PV technologies is the use of vertical fixed-tilt arrays. East-west vertical bifacial fixed-tilt PV arrays have competitive performance with south-facing panels in at high latitudes (Jouttijarvi et al., 2022; Pike et al., 2021), and are also being explored for agrivoltaic and building-integrated applications (Reker et al., 2022; Tahir and Butt, 2022).

It is generally acknowledged that horizontal single-axis tracked (HSAT) arrays require larger row spacing – or smaller GCRs – than fixed-tilt counterparts. For example, Berrian *et al.* cites GCRs of around 35% are typical for HSAT deployments whereas south-facing fixed-tilt GCRs are often >50% (Berrian et al., 2019). However, there is little existing literature which indicates how selection of the appropriate GCR varies with latitude, nor directly compares the performance of HSAT, south-facing fixed-tilt, and east–west vertical fixed-tilt arrays.

Many studies on row spacing and system design are limited to moderate latitudes around 15-40° either North or South of the equator, with few studies up to latitudes around 50°. For example, Baloch *et al.* examined the interplay of row spacing and mounting height on bifacial fixed-tilt and vertical PV arrays at 25°N, finding fixed-tilt arrays are more sensitive to mounting height than vertical arrays (Baloch et al., 2020). On the other hand, vertical arrays have been found to be more sensitive to shading loss due to row spacing compared to fixed-tilt arrays in a 32°N location (Appelbaum, 2016). Verissimo *et al.* compared how favorable PV array system designs vary with GCR for Brazil (Verissimo et al., 2020), Narvarte *et al.* compared tracking gain in Spain at 38°N (Narvarte and Lorenzo, 2008), and Al-Quraan *et al.* explored the interplay of PV array tilt and row spacing in Saudi Arabia (24°N) and Yemen (15°N) (Al-Quraan et al., 2022) .

Few studies have examined row spacing and system optimization in higher latitudes. Jacobson *et al.* conducted an extensive study on the effects of tracking type around the globe, considering cities with latitudes between 37°S and 64°N, and extrapolating to ±80° of the equator (Jacobson and Jadhav, 2018). However, this study was limited to monofacial systems and did not discuss how row spacing and land-use must vary by location. Khan *et al.* compared bifacial vertical arrays to monofacial equator-facing fixed-tilt arrays across the globe using a clear-sky model, and asserted that a 2 m row spacing should be used in practice (Khan et al., 2017). Pike *et al.* performed a seasonal comparison of monofacial and bifacial fixed-tilt and vertical PV module performance in Alaska, considering only single-module arrays (Pike et al., 2021). Frimannslund *et al.* compared the PV potential in the Arctic and Antarctic and discussed in a 78°N case study how the tilt of fixed-tilt arrays can be adjusted to better suit lower row spacing (Frimannslund et al., 2021).

On the bifacial side, research regarding bifacial system design configuration and challenges is becoming more common, however there has been little direct comparison between row spacing for bifacial vs monofacial technologies and still many recent publications on PV system design focus solely on monofacial systems (Bolinger and Bolinger, 2022; Jacobson and Jadhav, 2018).

In this paper we demonstrate how row spacing affects system performance for both monofacial and bifacial arrays, comparing south-facing fixed-tilt, HSAT, and east–west vertical configurations for North American locations at latitudes of 17°N up to 75°N. For each configuration and location, we quantify the inter-row shading and resulting system energy yield losses for GCRs between 0 and 1. We then provide latitude-optimal GCRs – and tilts, in the fixed-tilt case – for 5%, 10%, and 15% allowable annual energy yield loss due to inter-row shading.

2. Methodology

We selected 31 locations from across North American countries of Mexico, the United States, and Canada for PV row spacing characterization and optimization. Locations were selected to include a wide distribution in latitude (17°N to 75°N). Where possible, the locations represent multiple diffuse fractions within a given latitude range, with diffuse fraction defined as the annually-averaged ratio of diffuse horizontal irradiance (DHI) to global horizontal irradiance (GHI). Fig. 1 displays a map of diffuse fraction across the continent with our selected locations indicated by open circles. Environmental data was retrieved from the US National Solar Radiation Database (Sengupta et al., 2018) with the exception of latitudes >60°N, which were obtained from the Canadian Weather Energy Calculation database (Environment and Climate Change Canada, 2022). Typical Meteorological Year data was used for each location at an hourly resolution, including hourly-updating albedo values. A summary of pertinent environmental conditions for each location can be found in Table S1 of the Supplemental Information section.

We modelled the performance of the PV arrays depicted in Fig. 2 using our PV performance prediction software, DUET (Russell et al., 2022). DUET generates a detailed 3D model of the PV system including supportive structures and multiple rows. Optical calculations are then completed considering direct beam radiation, anisotropic diffuse sky radiation, and ground-reflected radiation by segmenting the modules, ground, and diffuse sky-dome into patches. A shading algorithm is implemented using a deterministic ray intersection method (Moller and Haines, 2002; Williams et al., 2005) to capture the effect of objects in the 3D scene. These shading results are combined with a 3D view-factor model to calculate two-dimensional front and rear irradiance profiles for each timestep (Coathup et al., 2023; Russell et al., 2022). Using environmental conditions and module electrical parameters, DUET translates these irradiance profiles into per-timestep cell current–voltage (*I-V*) curves through a temperature and irradiance-dependent single-diode model. Cell temperature is computed using the Sandia glass-cell-glass temperature model (King et al., 2004). Module *I-V* curves are constructed assuming the wiring diagram depicted in Fig. 2C, with 72-cells connected in series and parallel with three bypass diodes. The instantaneous module power is then scaled by the timestep (hour) to arrive at the module energy yield. Yearly total module energy yield, as analyzed in this work, is given by the summation of energy yield over all hourly timestamps. DUET has been validated against hourly field data from fixed-tilt and tracked bifacial PV systems to within a mean absolute error of <0.02 W/Wp (Russell et al., 2022; Stein et al., 2021).

To emulate utility-scale systems (Berrian et al., 2019), we calculated the energy yield of 1 central module in a 5-row array where each row contains 31 modules, as provided in Fig. 2A. To align with common deployment configurations, fixed-tilt and HSAT arrays are arranged in a 1-in-portrait (1P) configuration with module frames, purlins, and torque tubes as in Fig. 2B. Torque tube and purlin dimensions were chosen to align with the range of dimensions used in literature (Coathup et al., 2023; Deline et al., 2020). Vertical arrays are instead modelled in a 1-in-landscape (1L) configuration, as this is commonly used to reduce rear-side irradiance inhomogeneity (Valdivia et al., 2017; Wang et al., 2020). Modules were composed of either monofacial or bifacial silicon heterojunction (SHJ) solar cells with 20.2% efficiency and bifaciality of 96%. Module SHJ parameters, summarized in Table S2, were extracted

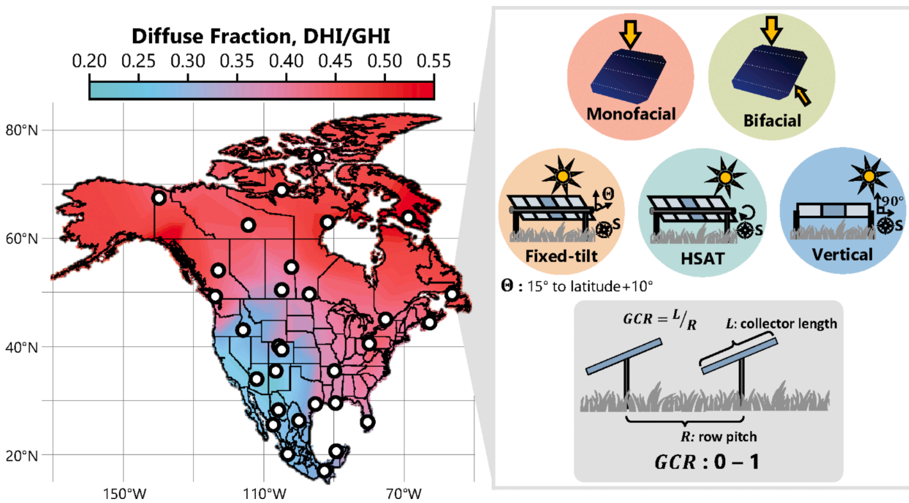

Fig. 1. A map of diffuse fraction across North American countries of Mexico, the United States, and Canada. White circles indicate the 31 locations studied. South-facing fixed-tilt systems are studied with tilts from latitude-tilt + 10° to a minimum tilt of 15°, in 5° increments. East-west HSAT systems studied have a range of motion covering ± 60°. Ground coverage ratios (GCRs) between 0 and 1 are studied for all illumination and mounting types, for both monofacial and bifacial modules. Depicted fixed-tilt, HSAT, and vertical arrays are simplified to show only 1 row containing 3–5 modules.

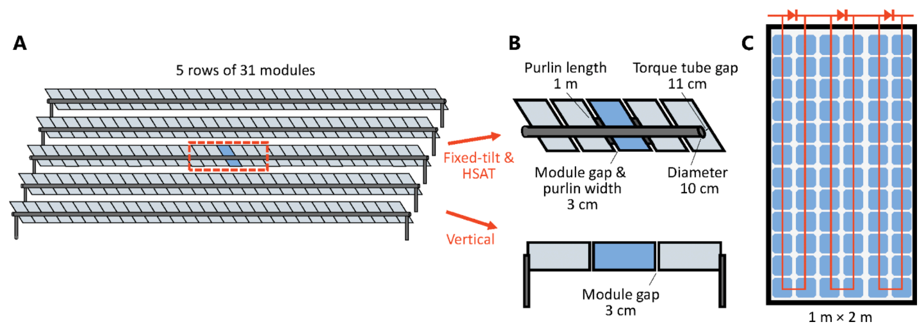

Fig. 2. (A) The full simulated PV array scene viewed from the rear-side for fixed-tilt, HSAT, and vertical arrays. Vertical modules are not tilted, as depicted. (B) Supportive structure dimensions. Purlins are simulated only around the central array module to reduce computation time in the case of fixed-tilt and HSAT arrays. (C) Framed-module layout and internal module wiring structure with bypass diodes indicated above the module.

using a validated drift–diffusion model for dual-sided textured photovoltaic devices described in Tonita et al. (2021). This model has been validated against measurements of SHJ devices (Tonita et al., 2021).

For this study, we calculated the effect of ground coverage ratios between 0 and 1 on monofacial and bifacial fixed-tilt and HSAT arrays, as visualized in Fig. 1. Vertical array GCR is only analyzed for the bifacial case, as the primary appeal of vertical PV arrays is for electricity generation in both morning and afternoon hours. GCR is defined by Equation (1) below:

$$GCR = \frac{L}{R} \tag{1}$$

where L is the collector length perpendicular to the row length and R is the row pitch. A GCR of 1 corresponds to the inter-row spacing where the horizontal gap between modules becomes zero if the module was to be rotated to horizontal tilt. A GCR of 0 corresponds to the case where only a single row in the array is considered, or the inter-row spacing between modules approaches infinity.

This study investigates full cell 1P arrays where L is the length of a single module (2.0 m) and 1L arrays where L is the instead the width of a single module (1.0 m), but results provided in terms of GCR will be generally applicable to other collector lengths, tier-number, and half-cut modules. For example, in Boulder, USA at 40°N, a bifacial full-cell 2-in-portrait (2P) fixed-tilt system with a tier gap above the torque tube of 0.2 m should have a GCR higher by 0.04 compared to the 1P case. Of a similar scale, a bifacial half-cut 1P fixed-tilt system should have a GCR higher by 0.03 compared to a full-cell system. Placing multiple electrically separated modules across the collector length, through either adding additional tiers or half-cutting cells, results in proportionally lower electrical mismatch loss and allows for marginally higher GCR. Due to comparably reduced irradiance inhomogeneity between tracked and fixed-tilt modules, the difference between 2P and 1P for HSAT system GCR need only be 0.01. Similarly, the difference between half-cut and full-cell HSAT system GCR is <0.01. Thus, calculated behaviour with GCR will apply to other collector lengths and half-cut cells.

For fixed-tilt arrays, we additionally varied the tilt angle, θ, from 10° greater than the latitude to a minimum tilt of 15° (in 5° increments) for all locations and all GCRs. A minimum limit of 15° is recommended in practice to reduce accumulation of dust and dirt, and to facilitate rain-based cleaning (Figgis et al., 2017; International Renewable Energy Agency, 2014). For the tilt, θ, we introduce a *latitude-tilt adjustment factor*, φ, for latitude, α, defined as:

$$\theta = \alpha + \varphi;\ \theta \geq 15^{\circ} \tag{2}$$

This latitude-tilt adjustment factor is introduced to emphasize the displacement from latitude-tilt during analysis. The ground clearance from the middle of modules to the ground, H, in meters was then adjusted according to the module tilt, θ:

$$H = \frac{L}{2}\sin\theta + 0.25 \tag{3}$$

We model our arrays with a modest minimum module ground clearance of 25 cm to align with tolerances of HSAT systems in the field, where H typically varies between 0.8 and 1.5 m (Ayala Pelaez et al., 2019; Berrian et al., 2019). Increasing this minimum ground clearance to 1.0 m for fixed-tilt, HSAT, and vertical arrays decreases the optimal GCR in Boulder by 0.01, 0.02, and 0.01, respectively. For the HSAT systems modelled in this work, H is set to 1.12 m where θ in Eq. (3)

equates to the tracking bounds of ± 60° tilt. In addition, a conventional backtracking algorithm is employed for all HSAT arrays (Lorenzo et al., 2011). In this algorithm, HSAT module tilt is adjusted away from direct-beam normal incidence to eliminate direct shading between adjacent rows. For further information, a summary of all system inputs is provided in Table S2.

3. Results

3.1. Influence of ground coverage ratio on optimal tilt in a fixed-tilt system

The interplay of GCR and tilt of fixed-tilt modules is an optimization problem. As GCR increases (or inter-row spacing decreases), energy yield loss due to shading for a given module tilt becomes more and more substantial. For higher GCR values, a lower tilt enhances energy yield by reducing row-to-row shading loss, despite additional cosine losses (Al-Quraan et al., 2022; Rehman et al., 2020; Van Schalkwijk et al., 1997). Fig. 3A displays how changing the tilt of a fixed-tilt array away from latitude-tilt by φ affects the bifacial energy yield of a central module located in a 5-row array in Boulder, USA. For a given GCR, the tilt adjustment factor providing the highest energy yield, φ_{opt}, is displayed by a star. To the right of the star shading losses dominate, while to the left of the star cosine losses dominate. As GCR increases from 0 to 1.0, the optimal tilt of the module moves from latitude + 5° to the minimum allowed tilt of 15°, or tilt adjustment factor of −25° for Boulder, Colorado. One can also observe on this plot the substantial influence GCR has on module energy yield, particularly for GCRs > 0.5. From a GCR of 0 to 0.5 the energy yield drops by 9%. From a GCR of 0.5 to 1.0, the energy yield drops by 50%.

The latitude of the fixed-tilt array also affects the optimal tilt adjustment factor. Here, we combine the effect of GCR, latitude, and module tilt for latitudes between 20 and 75°N. Fig. 3B and 3C display the optimal tilt adjustment factors as a function of GCR for different latitude bins for monofacial and bifacial modules, respectively. Locations are binned and averaged by latitude within the error bounds indicated in the legend, with 2–4 locations per bin. As a contrast to previous findings which suggest using tilt adjustment factors between −5° to −15° (Calabro, 2013; Lewis, 1987; Qiu and Ruiffat, 2003), we find that the appropriate latitude-tilt adjustment factors vary between +7° to −60° when a wide range of latitudes and GCRs are considered. At all latitudes, shifting towards higher GCR favours shallower tilt angles in order to limit inter-row shading loss, as seen in Boulder in Fig. 3A. For GCR >0.7, the minimum tilt of 15° should be used at all latitudes. In higher latitude locations, the tilt adjustment factor is significantly more sensitive to GCR due to lower average solar elevation. For example, when changing GCR from 0.3 to 0.4, the optimum tilt adjustment factor for a monofacial fixed-tilt module at 17°N decreases by 0.4°, while the optimum tilt for this same module located at 75°N decreases by 14°.

Additionally, whether the modules are monofacial or bifacial can have a slight impact on the choice of optimal module tilt. For GCRs >0.5 bifacial and monofacial optimal tilts are about the same since the irradiance contribution from rear-incident light is proportionally small. However, when GCR decreases below 0.5, there is more opportunity for diffuse, reflected, and occasionally direct-beam radiation to reach the rear-side, increasing the relative contribution of row shading on the front side. The balance between these two effects leads to higher optimal tilt angles for bifacial arrays than for monofacial arrays at lower GCRs. For the scenario of no neighbouring rows (GCR = 0), bifacial module tilts should be on average 3° steeper than monofacial tilts for all locations. For a moderate GCR of 0.35, bifacial modules optimum tilts are between 0 and 3° steeper than monofacial tilts, and 1° steeper on average. However, this difference depends on the bifaciality of the technology used and will be smaller for modules with bifaciality <96%; thus, in most cases, the optimum tilt of monofacial and bifacial fixed-tilt arrays can be assumed to be about the same.

All fixed-tilt results that follow are with the tilt set to the appropriate optimum value.

3.2. Influence of ground coverage ratio on inter-row losses

Next, we quantified the sensitivity of annual module energy yield to changes in GCR for fixed-tilt, HSAT, and vertical arrays. Fig. 4 displays the results for a bifacial module in a fixed-tilt array (4A), HSAT array (4B), and vertical array (4C). The inter-row energy yield loss, or the amount of energy yield that is lost in comparison to an infinite row spacing case (GCR = 0), is plotted on the y-axis. Thus, this inter-row energy yield loss includes losses associated with direct-beam inter-row shading, diffuse-sky masking from adjacent rows, and, in the case of HSAT modules, the cosine losses associated with backtracking to minimize shading.

There is a substantial difference in the impact of GCR on fixed-tilt, HSAT, and vertical arrays for low-to-moderate latitudes. For example, at our lowest latitude case of Tuxtla Gutiérrez, Mexico at 17°N, 5% inter-row shading loss occurs at a GCR of 0.68 for fixed-tilt arrays compared to a GCR of 0.32 for HSAT arrays and 0.16 for vertical arrays. Lower latitude locations with fixed-tilt arrays have a wide range of GCRs with inter-row energy yield loss <5%. At our highest latitude location in Resolute Bay, Canada at 75°N, the difference between fixed-tilt and HSAT array GCR causing 5% loss is only 0.02. As a contrast, the energy yield lost to inter-row shading effects steeply increases with GCR for vertical arrays, with less sensitivity to latitude than fixed-tilt and HSAT cases. To achieve 5% shading loss, vertical array GCR must be <0.20 for all locations.

Monofacial module trends are similar to the bifacial trends plotted, only shifted by 11% on average towards higher GCR for equivalent

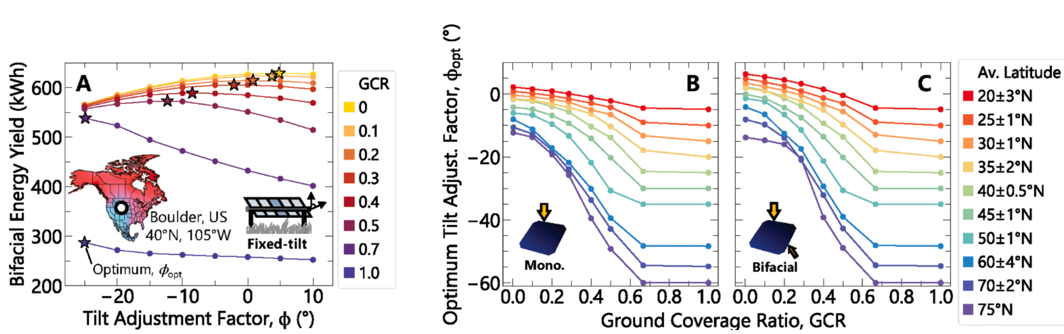

Fig. 3. (A) The bifacial energy yield of a central fixed-tilt module in a 5-row PV array as the tilt adjustment factor, ϕ, is varied from −25° to + 10° for Boulder, USA. A tilt-adjustment factor of zero corresponds to latitude-tilt. Stars indicate the interpolated point of highest energy yield, corresponding to the optimal tilt-adjustment factor for a given GCR. The optimum tilt-adjustment factors for different GCRs and latitudes is plotted in (B) for monofacial and (C) bifacial south-facing fixed-tilt systems. Data for the latitudes provided is an average of 2–4 locations, located within the given latitude-bin. The constraint of a minimum tilt of 15° causes plateaus in the adjustment-factor for GCRs > 0.7.

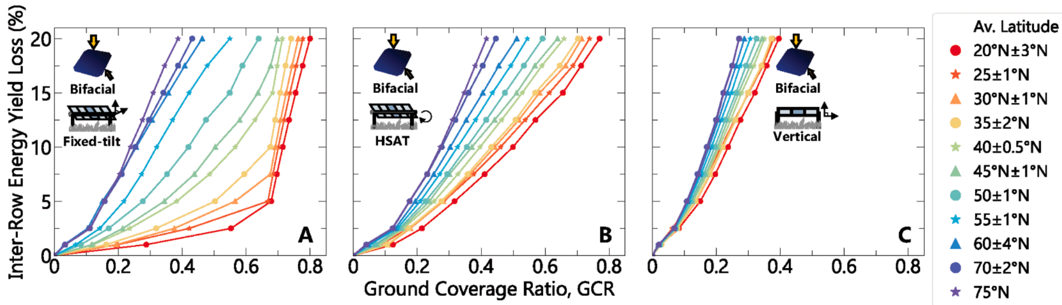

Fig. 4. The impact of GCR on the total amount of energy yield lost to inter-row effects of direct-beam shading and diffuse-sky radiation masking for all 31 North American locations for bifacial (A) fixed-tilt systems, (B) HSAT systems, and (C) vertical systems.

annual relative inter-row losses. Further comparison between bifacial and monofacial results is presented and discussed in the following section.

3.3. Latitude trends

To further elucidate GCR trends between 17 and 75°N, we consider the requirements for each system configuration to perform with an accepted inter-row energy yield loss of 5%. Fig. 5 displays the GCR for each of our 31 locations that results in 5% inter-row losses parsed by latitude and diffuse fraction for fixed-tilt arrays (5A), HSAT arrays (5B), and vertical arrays (5C). Inset in each of these plots is a linearly-interpolated geographical map displaying the GCRs.

While there is a correlation between diffuse fraction and latitude, with a Pearson correlation coefficient of r = 0.71, we find that GCRs are more strongly influenced by latitude than diffuse fraction. This can be seen by the dominating shift of GCR from low to high latitude. Locations at similar latitudes but different diffuse fractions have an average standard deviation in GCR of just ±0.02.

Fixed-tilt GCRs achieving only 5% inter-row energy yield loss span between 0.14 and 0.68 from 75°N to 17°N for bifacial modules, while HSAT GCRs range between 0.18-0.32 and vertical GCRs between 0.10-0.16. To achieve a given relative inter-row energy yield loss, HSAT arrays generally require lower GCR values than fixed-tilt arrays, especially for lower latitude locations. However, in Canada and Alaska, spacing requirements are similar for the two configurations, and HSAT arrays even allow narrower row spacing than fixed-tilt arrays for latitudes >55°N. This is more easily seen in Fig. 6, which shows the latitude-trends of GCRs achieving inter-row energy yield losses of (A) 5%, (B) 10%, and (C) 15%. These higher inter-row loss scenarios would apply for PV deployments where land is limited and modules must be spaced closer together to achieve desired yields.

Trendlines for monofacial and bifacial modules in fixed-tilt, HSAT, and vertical arrays are plotted in Fig. 6. Fixed-tilt behaviour is fit to an S-curve, with the following equation:

$$GCR = \frac{P}{1 + e^{-k(\alpha - \alpha_0)}} + GCR_0 \tag{4}$$

where P, k, α_0, and GCR_0 are fitting parameters and α is the latitude. All fitting parameters for the subplots in Fig. 6 are provided in Table 1, allowing for the computation of appropriate GCR for any latitude between 15-75°N. For the HSAT and vertical array linear regression parameters, m is the slope, while b represents the y-intercept. The standard deviation between the fitting functions and calculated data points is ±0.03 for fixed-tilt arrays and ±0.01 for HSAT and vertical arrays.

Fixed-tilt array GCR exhibits S-curve behaviour, with low-latitude GCR plateauing due to a minimum tilt of 15° and high-latitudes experiencing diminishing returns on further increases in row spacing. HSAT and vertical array behaviour is fit by a linear regression, with GCR decreasing linearly towards higher latitudes. Vertical arrays only show a marginal dependence on latitude, leading to a simple linear fit. While HSAT behaviour is more sensitive to latitude, HSAT trends are different than south-facing fixed-tilt arrays due to shade-reduction via backtracking. For a given GCR, HSAT arrays at higher latitudes will backtrack more often than lower latitude arrays, resulting in higher cosine losses. Therefore, higher latitude systems require lower GCR to maintain a specific inter-row loss.

For all depicted inter-row shading loss scenarios, HSAT GCRs and fixed-tilt GCRs are comparable for latitudes >55°N. This means that land requirements for deployments in the North will be similar for HSAT and fixed-tilt arrays. Land-usage should not dictate whether single-axis

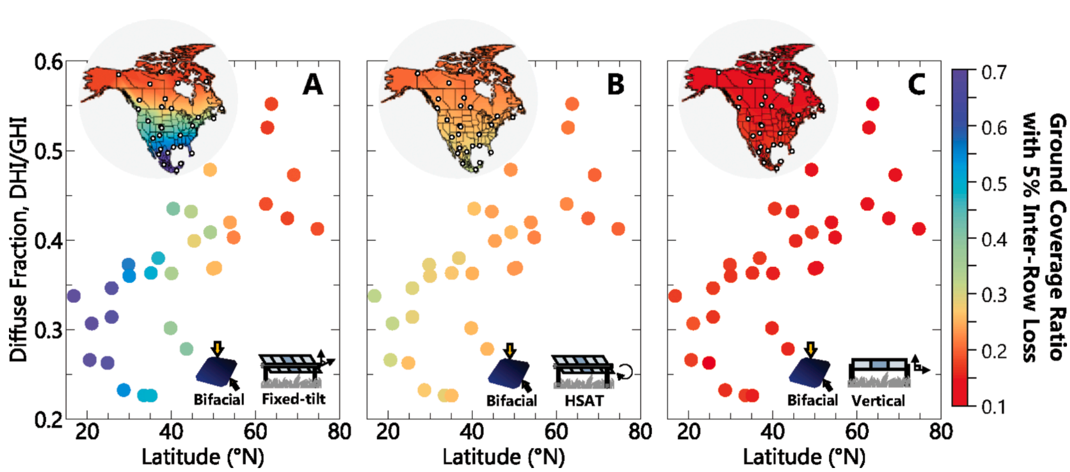

Fig. 5. The GCR giving a 5% inter-row spacing energy yield loss for all 31 locations as a function of latitude and diffuse fraction for (A) bifacial fixed-tilt systems, (B) bifacial HSAT systems, and (C) bifacial vertical systems. Inset maps show the GCR linearly interpolated across Mexico, the United States, and Canada.

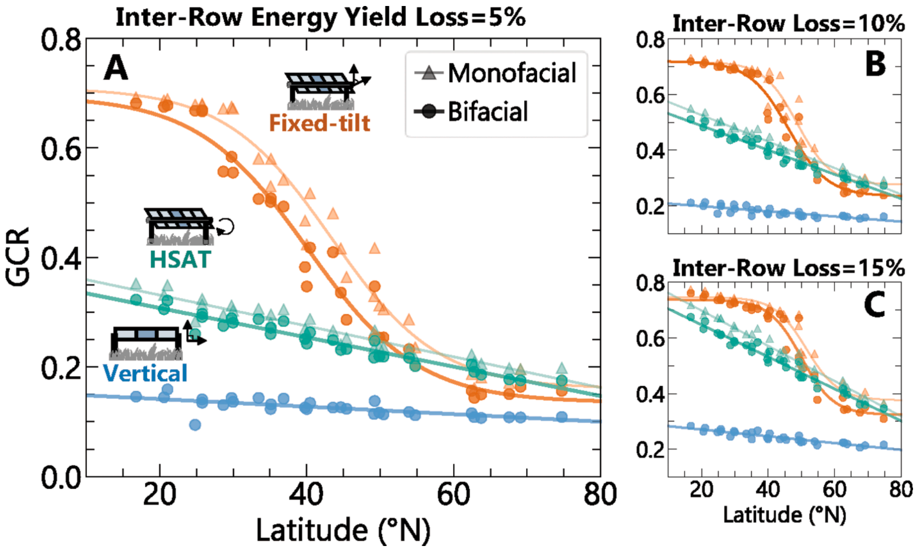

Fig. 6. GCR latitude-trends for (A) inter-row energy yield shading loss ('inter-row loss') of 5%, (B) 10%, and (C) 15%. Fixed-tilt data is given in orange, while HSAT data is given in teal and vertical data in blue. Monofacial and bifacial system results are additionally provided by lighter triangles and darker circles, respectively, with deviations between bifacial and monofacial performance emphasized by the high module bifaciality of 96%. The appropriate GCR for any latitude > 15°N can be extracted from these plots, with fit parameters summarized in Table 1. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 1
Fitting Parameters for Calculating Ground Coverage Ratio.

		Fixed-tilt				HSAT		Vertical	
		P	*k* (1/°)	α_0 (°)	GCR_0	*m* (1/°)	*b*	*m* (1/°)	*b*
Inter-row energy yield loss 5%	Bifacial	−0.560	0.133	40.2	0.70	-2.68×10^{-3}	0.361	-6.87×10^{-4}	0.155
	Monofacial	−0.550	0.138	43.4	0.71	-2.82×10^{-3}	0.388	–	–
Inter-row energy yield loss 10%	Bifacial	−0.485	0.171	46.2	0.72	-4.37×10^{-3}	0.575	-1.23×10^{-3}	0.257
	Monofacial	−0.441	0.198	48.7	0.72	-4.76×10^{-3}	0.621	–	–
Inter-row energy yield loss 15%	Bifacial	−0.414	0.207	49.9	0.74	-5.76×10^{-3}	0.762	-1.93×10^{-3}	0.357
	Monofacial	−0.371	0.208	51.5	0.75	-6.33×10^{-3}	0.825	–	–

trackers are deployed in the North, instead decision-makers can focus on other relevant factors such as tracking cost, feasibility, and energy yield gain.

For low-to-moderate latitudes, HSAT array inter-row shading loss is more sensitive to changes in GCR than fixed-tilt systems. For example, in Miami, USA at 26°N, the GCR for bifacial HSAT arrays is lower than fixed-tilt arrays by 0.13 under the 15% loss scenario, but this discrepancy nearly triples to 0.37 under the 5% loss scenario. Contrary to the high-latitude case, low latitude decision-makers who wish to minimize inter-row shading losses must consider land-use as a key factor when choosing between tracking and fixed-tilt systems. Vertical system GCR may reach up to 0.29 when shading loss is more tolerable at low latitudes, as in the 15% shading loss scenario in Fig. 6C. However, low GCRs <0.20 are necessary for all latitudes when limiting shading loss to 5%. This suggests that vertical PV arrays may need to tolerate higher shading losses to be more feasible in practice compared to south-facing fixed-tilt and HSAT arrays.

Whether the array is bifacial or monofacial additionally affects the latitude-optimal GCRs. As shown in Fig. 6, the trends between bifacial and monofacial modules for fixed-tilt and HSAT arrays follow the same behaviour, with bifacial module GCRs shifted lower on average by 0.03 than monofacial, since larger row spaces leads to higher rear irradiance. The appropriate GCR for any configuration at any latitude >15°N can be extracted from these plots or calculated using the fitting parameters provided in Table 1.

4. Discussion

While this paper discusses the interplay of inter-row energy yield loss on GCR for tracking/fixed-tilt/vertical systems and bifacial/monofacial modules, the appropriate selection of system configuration is a complex economic objective function, and further design optimizations should be considered. For example, PV arrays can be further optimized by accounting for non-flat terrain (Al-Quraan et al., 2022), considering the effects of wind-based module cooling caused by array layout (Glick et al., 2020), alternating panel inclination between neighboring rows (Kafka and Miller, 2020), varying mounting height (Baloch et al., 2020), and using row-specific tracking algorithms for HSAT arrays (Daly and Abbaraju, 2018). In addition, other non-conventional array mounting configurations may be of benefit. For example, high density, shallow ground-mounted array layouts with little spacing between array rows have been recently emerging onto the market, claiming land-use reductions ~30% (Carroll, 2021; GameChange Solar, 2023; Jurchen Technology, 2023).

Here, we presented the sensitivity of inter-row energy yield shading loss to GCR by comparing to the energy yield of a module with no neighbouring rows (GCR = 0). In some cases, it might be more beneficial to instead consider energy density, as suggested by Verissimo et al. (2020). This could be of more use when, for example, available land for PV is limited and module costs are low. Figure S1 in the supplemental information shows a plot of energy yield density as a function of GCR for the 31 locations considered in this work. For HSAT arrays, energy yield density continues to increase towards GCRs of 1.0, despite increased shading losses. If the sole objective is to maximize the energy yield of a given plot of land regardless of module cost, HSAT rows can be placed with little to no gap between rows, since losses associated with near constant backtracking are offset by the sheer number of modules on site. However, this is not a practical site layout in most circumstances as space must be left for operation and management practices, and modules spend most of the year backtracking to tilts < 15°. Fixed-tilt energy density, on the other hand, peaks for GCRs between 0.5 and 0.7 for latitudes between 17°N to 75°N. Even with negligible module cost, GCRs beyond these peaks are unlikely to be economically feasible, as reducing row spacing further results in an overall loss to energy yield; the benefit gained by fitting more modules into a given space is very literally overshadowed by shading losses if a minimum 15° tilt is maintained to avoid soiling. Eliminating the minimum tilt requirement of 15° can allow for non-conventional shallow-tilt array layouts with high energy yield density at GCRs of 1 (Carroll, 2021; GameChange Solar, 2023; Jurchen Technology, 2023). Unlike their equator-facing fixed-tilt counterparts, east–west facing vertical panels do not experience an energy yield density peak for GCRs up to 1; it is possible to have GCRs >1 to increase the energy yield density of the PV array (Khan et al., 2017; Riaz et al., 2021; Tahir and Butt, 2022). In agrivoltaic applications, the effect of vertical PV row spacing on crop yield must also be considered, with certain crops being more shade-tolerant or shade-sensitive (Riaz et al., 2021; Tahir and Butt, 2022). In all cases, incorporating the effect of GCR on both module energy yield and energy yield density will be necessary to help inform PV array design.

PV arrays were composed of full-cell silicon technology for the presented GCR and energy yield analysis. The power output of a module is generally proportional to the amount of incident irradiance, however, some technologies may be more shade-tolerant. For example, both half-cut cells and cadmium telluride (CdTe) technologies have demonstrated a lower sensitivity to shading than full-cell silicon modules, resulting in tracking schemes that reduce or even eliminate backtracking hours for overall improved performance (Cheein et al., 2021; Ngan et al., 2013). In this case, restrictions on appropriate GCR values as a function of latitude will be lessened, and closer row spacing could be used.

The analysis we have presented has been conducted for North American locations, but covers a wide range of operating conditions, including diffuse fractions between 0.23 and 0.55, average GHI-weighted ambient temperatures of −4°C to 31°C, average GHI-weighted albedos between 0.10 and 0.65, and city elevations between 1 and 1600 m. These location parameters are summarized in Table S1. Including locations across other continents and in the southern hemisphere in this analysis would further the range of operating conditions and contribute to more scatter in the presented trends. Since latitude was the dominating factor effecting inter-row energy yield loss, our results should provide an estimate of the performance of equivalent PV arrays across the globe.

5. Conclusion

Traditional guidelines for determining PV array layouts were developed for monofacial fixed-tilt equator-facing systems at low-to-moderate latitudes, and no longer suit well the expanding PV market, which has been progressing toward bifacial technologies, tracked systems, higher latitudes, and land-constrained areas. Old approaches, like the winter solstice rule, are insufficient to capture the nuance of deployment planning, where practical and economical constraints vary widely both geographically and temporally. In this work, we have quantified the row spacing for tracked, fixed-tilt, and vertical arrays with both bifacial and monofacial technologies in 31 locations from 17–75°N with acceptable inter-row shading losses of 5–15%. We generalize our results to an arbitrary collector area by presenting results in terms of ground coverage ratio. GCR varies widely between 0.15–0.68 for fixed-tilt systems compared to 0.17–0.32 for HSAT systems, both with a strong latitude-dependence. Similarly, the optimal tilt of fixed-tilt arrays varies widely from 7° above latitude-tilt to 60° below latitude-tilt, depending on the latitude and GCR. Vertical systems are less sensitive to latitude, with GCR varying from 0.10 to 0.16 between 17-75°N. We also find that it is reasonable to approximate the row spacing of bifacial arrays as equivalent to monofacial arrays, with bifacial modules of 96% bifaciality requiring GCRs lower by 0.03 on average than monofacial modules. We demonstrate that latitude is a stronger driver of inter-row energy yield shading losses than diffuse fraction, and present formulae for calculating the appropriate row spacing of a PV array for any latitude between 15–75°N. Our results provide updated guidelines for PV deployment system design that better suit the expanding PV sector.

6. Data and code availability

The data supporting the findings of this study are available in the main text and supplemental information and are additionally available upon request from the corresponding author. Details regarding the code used in this paper are published in Russell et al. (2022).

CRediT authorship contribution statement

Erin M. Tonita: Conceptualization, Methodology, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Annie C.J. Russell:** Conceptualization, Methodology, Software, Validation, Investigation, Writing – review & editing. **Christopher E. Valdivia:** Conceptualization, Writing – review & editing, Supervision. **Karin Hinzer:** Conceptualization, Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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The University of Ottawa is located on the unceded territory of the Anishinaabe Algonquin Nation.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.solener.2023.04.038.

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